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quantum information and photonics

MUQIP

Integrated Photonics Signal Processing

Lab 1

Waveguides

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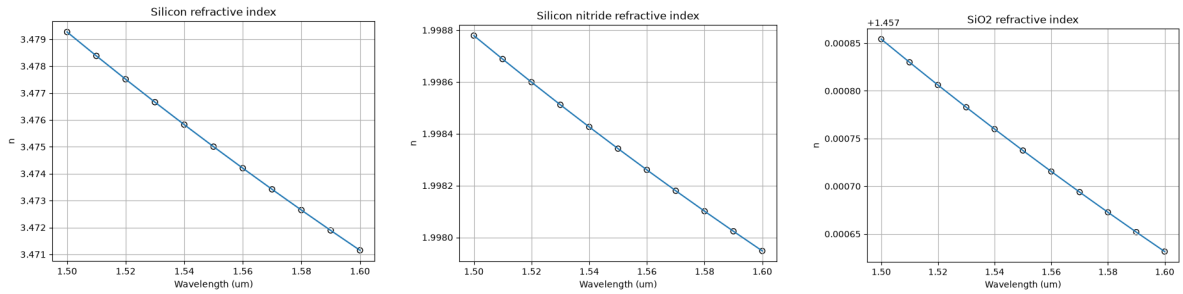
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Learning outcome 1 – Effective index of a waveguide

Results.

Part 1. Obtaining the characteristics of the materials to work with

The Tidy3D library has been used to obtain the characteristics of Si, Si_3N_4 and SiO_2. The plots in figure 1 show their refractive index on the vertical axis and the corresponding wavelength on the horizontal axis. We can observe that Si offers the highest refractive index and SiO_2 the lowest. Furthermore, we can observe that the refractive index of SiO_2 varies very little with the wavelength in comparison with the other two materials.



(a) Si. **(b)** Si_3N_4. **(c)** SiO_2.

Figure 1: Refractive index of the materials versus wavelength.

Part 2. Cross-section definition

Next, using the GDSFactory–Tidy3D mode solver, we define the waveguide we are going to work with. Specifically, in this case a deep waveguide has been configured, with a silicon dioxide cladding and a silicon nitride core. It has been configured with a height of 300 nm and a width of 1.2 µm. The simulations are intended to be performed at a wavelength of 1.55 µm, so this is also configured, together with other necessary simulation parameters. In figure 2 the refractive index distribution over the waveguide cross-section (x–y plane) can be observed.

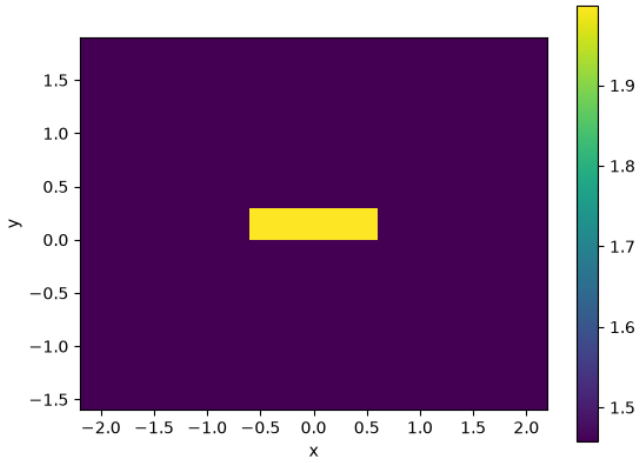


Figure 2: Refractive index of the waveguide cross-section.

Part 3. Simulations

Fewer than 4 modes are expected to be found, so when the waveguide was configured it was set to search for a maximum of 4 solutions. Using `deep_waveguide.n_eff` we can obtain the effective index of each mode, the ones found being those shown in table 1.

Table 1: Effective index of each mode (real part).

Mode	n_{eff}
0	1.6052
1	1.5283
2	1.4507
3	1.4333

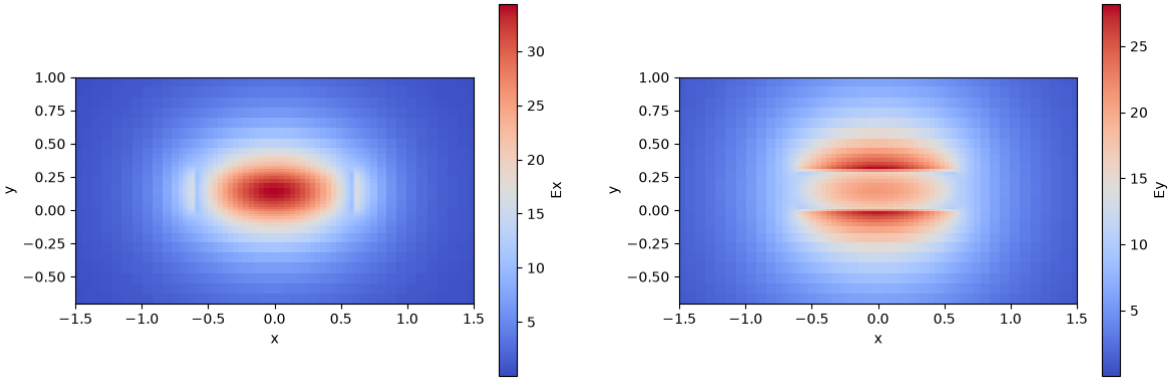
A mode is guided when its n_{eff} is lower than the refractive index of the core and higher than that of the cladding ($n_{\text{clad}} \approx 1.457$), so the guided modes are modes 0 and 1. To find out which type of mode we have, we can use `deep_waveguide.fraction_te` and `deep_waveguide.fraction_tm`, whose results are shown in table 2. When the value approaches 1, it is because the mode belongs to that type, so the first guided mode is of TE type and the second one is of TM type.

Table 2: TE and TM fraction of each mode.

Mode	TE fraction	TM fraction
0	0.995	0.005
1	0.010	0.990
2	0.963	0.037
3	0.048	0.952

Finally, figures 3a and 3b show the field for the two guided modes. Thanks to these plots it is possible to evaluate, among other things, the confinement, that is, how much field is inside the core and how much has "spilled" outside. In the case of TE-type modes, the electric field points mostly horizontally, so we plot the E_x field. For the TM mode we have the opposite case, where the field points mostly vertically, so we plot the E_y field.

In figure 3 we can observe how mode 0, the TE-type one, is very confined inside the waveguide, while mode 1, the TM-type one, is significantly less confined, with the field spreading out through the top and bottom walls. As the TE type has its field mostly pointing horizontally, its losses are concentrated on the left and right walls, so defects in these have a great impact on this type of mode. Analogously, the TM type has its field mostly pointing vertically, so its losses are concentrated on the top and bottom walls, and defects in these have a great impact on this type of mode.



(a) Mode 0 (TE), E_x.

(b) Mode 1 (TM), E_y.

Figure 3: Field of the guided modes.

Learning outcome 2 – Width dependence

Results.

The sweep has been configured to start at 0.6 µm and end at 1.6 µm, with steps of 0.1 µm, with the aim of studying how the modes evolve for different widths. In figure 4 we can observe the results. In general, we observe how the effective index increases as the width increases, although it is possible to observe how the TE modes (those with red points) undergo a slightly more pronounced increase, which is probably due to the fact that, as mentioned above, the field of these modes is mainly horizontal, so changes in width are expected to affect them more.

We can also observe how, with widths smaller than 1.3 µm, we only find one mode per type, that is, one TE mode and one TM mode. However, from a width of 1.3 µm, mode 2, of TE type, starts to be guided, and mode 3, of TM type, is close to being guided at a width of 1.6 µm.

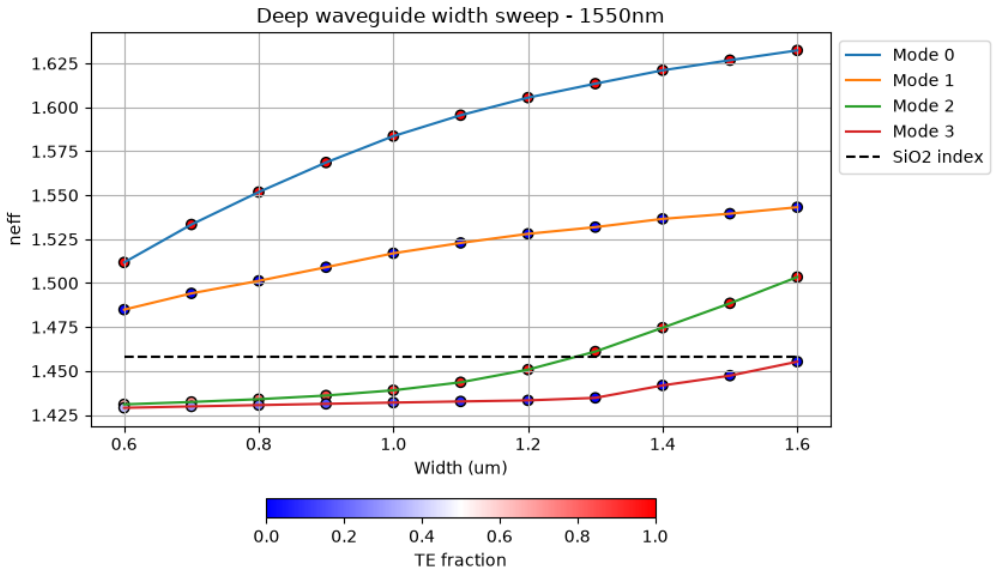


Figure 4: Effective index of the modes versus waveguide width.

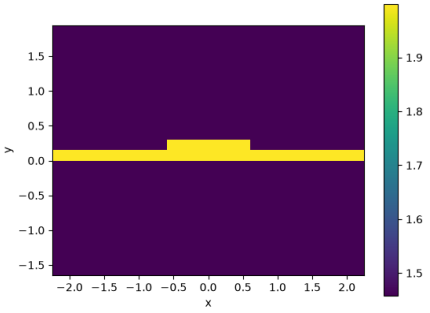
Learning outcome 3 – Wavelength behavior

Results.

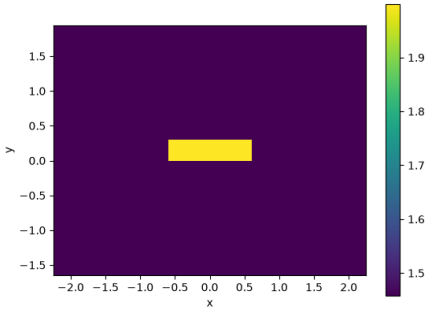
In the deep waveguide, which we can see in figures 5b and 5d, we can observe how, as the wavelength increases, the effective index decreases for all modes, slightly more markedly for modes 0 and 1. We can observe how the jump in effective index is relatively large between modes 0, 1 and 2, while it is considerably smaller between modes 2 and 3.

In the case of the shallow waveguide, unlike the deep one, the effective index of the 4 modes decreases with a similar slope. We can also see how the jump in effective index between mode 0 and mode 1 is relatively large, while the jumps between the rest of the modes are much smaller.

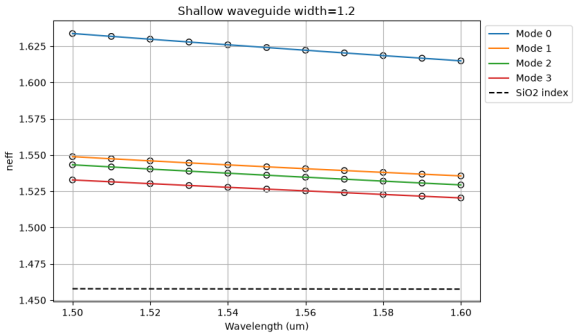
Comparing the two types of cross-section, we can observe how in the shallow type we obtain the highest effective index, slightly above 1.625, while in the deep type it stays below this mark. In general, the effective index of all the modes is higher in the shallow type.



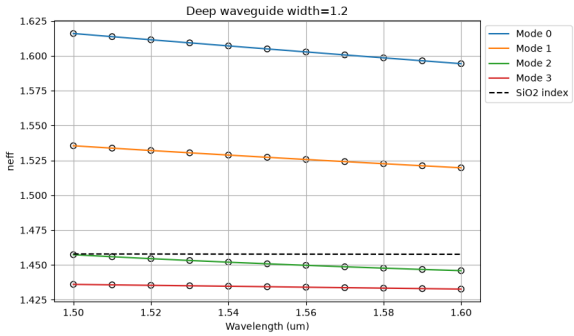
(a) Shallow, cross-section.



(b) Deep, cross-section.



(c) Shallow, n_{eff}.



(d) Deep, n_{eff}.

Figure 5: (a, b) Cross-section at 1.55 µm and (c, d) effective index of the modes versus wavelength.

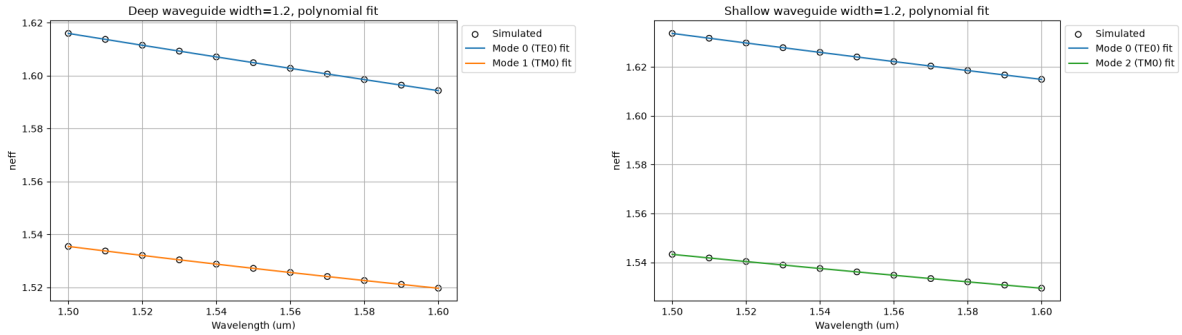
Learning outcome 4 – Waveguide compact model

Results.

To find the second-degree polynomial coefficients, we started by choosing $\lambda_0 = 1.55\,\mu\text{m}$, as it is the central value of the swept range. Next, we obtain the n_{eff} of the TE_0 and TM_0 modes from the simulations of learning outcome 3. It should be noted that in the shallow waveguide the TM_0 mode is mode 2, instead of mode 1 as in the deep waveguide, since the TE fraction of mode 1 is 0.706, while that of mode 2 is 0.175. To obtain n_1, n_2 and n_3 we fit the polynomial with `np.polyfit()`, which gives us the coefficients that can be seen in table 3.

Once we have obtained the coefficients, we can obtain the group index $n_g = n_1 - n_2\lambda_0$ and the dispersion parameter $D = -2\lambda_0 n_3/c$ and, making sure that we convert to the correct units, we obtain the values that can be seen in table 3. This process has been repeated for each mode and for each type of waveguide.

In figure 6 we can observe, as circles, the simulated values, and as a line, the fitted polynomial, that is, the values calculated by the model. As can be observed, the model fits the simulated data perfectly, which was the expected result.



(a) Deep. **(b)** Shallow (slab 150 nm).

Figure 6: Simulated n_{eff} of TE_0 and TM_0 and second-degree polynomial fit.

Table 3: Fit coefficients and compact model parameters at $\lambda_0 = 1.55\,\mu\text{m}$.

Waveguide	Mode	$n_1 = n_{\text{eff}}$	n_2 (1/µm)	n_3 (1/µm^2)	n_g	D (s/m^2)
Deep	TE_0	1.6049	−0.2162	0.0974	1.9400	-1.007×10^{-3}
	TM_0	1.5272	−0.1580	0.1538	1.7720	-1.591×10^{-3}
Shallow	TE_0	1.6241	−0.1882	0.0957	1.9158	-9.895×10^{-4}
	TM_0	1.5361	−0.1387	0.0980	1.7510	-1.013×10^{-3}

Analyzing the results, we can observe that in the case of the deep waveguide we obtain higher values for TE_0 for both the effective index n_{eff} and the group index n_g, while the absolute value of the dispersion parameter D is smaller. This behavior is consistent with what was observed in learning outcome 1, where the TE_0 mode also offers a higher effective index and more confinement than TM_0. Analogously, we can observe how the results for the shallow waveguide are similar, with higher n_{eff} and n_g for TE_0, and a smaller $|D|$.

Comparing the two types of cross-section, we see how the shallow one offers a higher n_{eff} and a smaller absolute value of D, and its group index n_g is slightly lower. Interpreting these results, we can state that the waves will travel more confined in the high-index material (core and slab) in the shallow waveguide, and since its group index is lower, the signal will travel faster.

Assignment

Results.

Now we repeat learning outcomes 2 and 3 changing the core material to silicon (Si) and also adjusting the height to 220 nm.

Part 1. Width dependence

First, we study how the effective index evolves with respect to the core width, studying the range from 0.3 µm to 1 µm. The results can be observed in figure 7. We can observe how, as in the previous case (learning outcome 2), the effective index increases as the width does, although the curves are slightly different. We can observe that below a width of 0.4 µm we have a single-mode waveguide, with a single mode per polarization, that is, a single TE and a single TM. For this reason, a width of 0.35 µm will be used for the wavelength dependence analysis, since it is required to do it using the waveguide in a single-mode region.

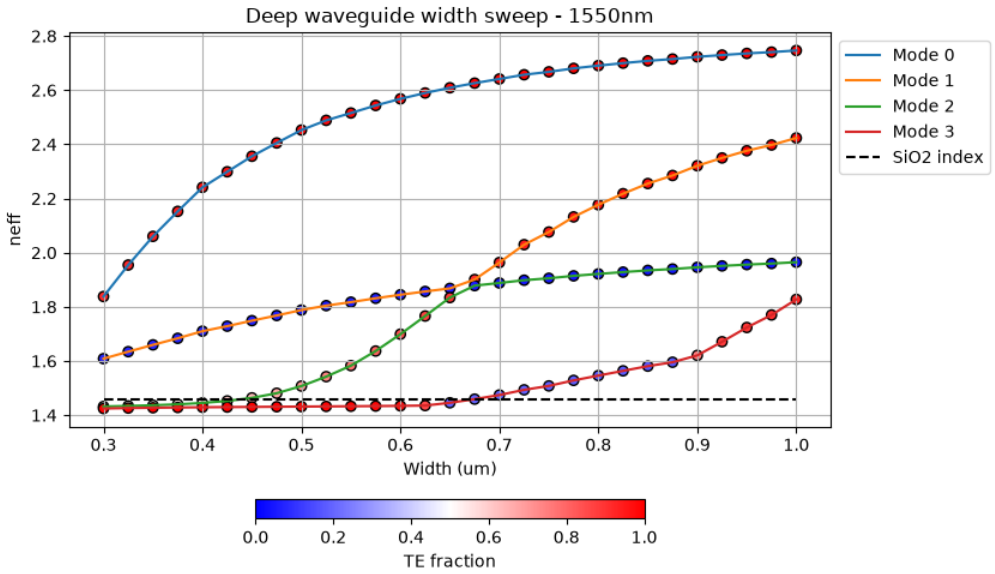
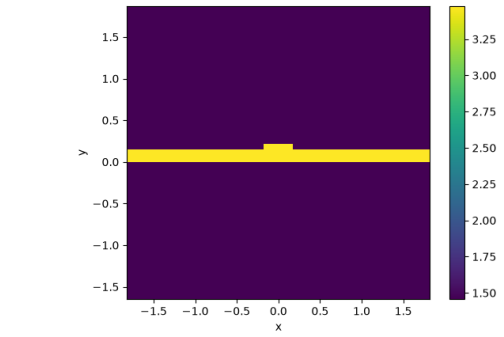


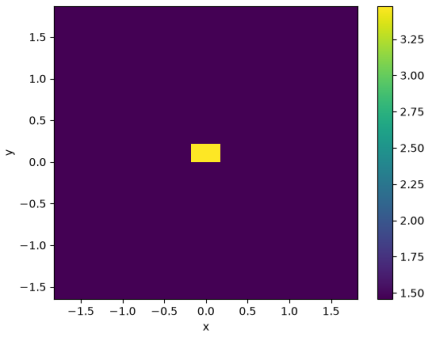
Figure 7: Effective index of the modes versus waveguide width (SOI, 220 nm).

Part 2. Wavelength behavior

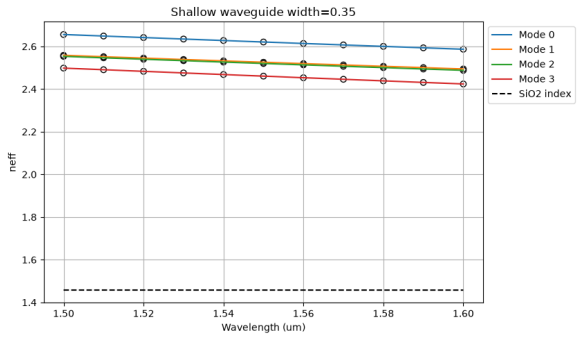
For the wavelength behavior analysis, a width of 0.35 µm has been used, as mentioned previously, and it has been studied from 1.5 µm to 1.6 µm. In the case of the deep waveguide (figure 8d), we can observe that in the guided modes the effective index decreases with respect to the wavelength. The decrease is similar between the two guided modes, that of mode 0 being slightly larger. As for the shallow waveguide (figure 8c), we can observe that all the modes are relatively grouped together, and they all decrease in a similar way. Comparing the types of waveguide, we can observe how the shallow one presents modes with a higher effective index with respect to the deep waveguide.



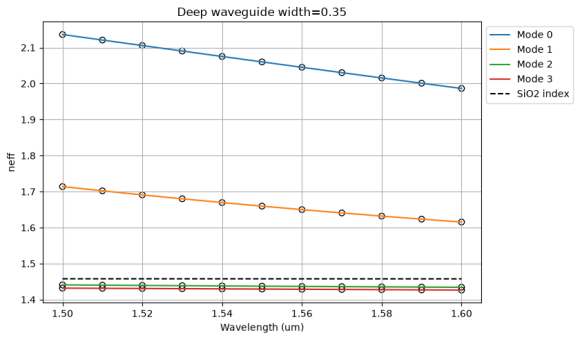
(a) Shallow, cross-section.



(b) Deep, cross-section.



(c) Shallow, n_{eff}.



(d) Deep, n_{eff}.

Figure 8: SOI, $W = 0.35\,\mu\text{m}$: (a, b) cross-section at $1.55\,\mu\text{m}$ and (c, d) effective index of the modes versus wavelength.

Part 3. Comparison between Si_3N_4 and SOI

We close this question by comparing the differences observed between the waveguide with a Si_3N_4 core and SiO_2 cladding and the waveguide with a Si core and SiO_2 cladding, known as Silicon-On-Insulator (SOI) technology.

The first difference is the core index contrast, about 2.00 for the Si_3N_4 case and about 3.48 for the Si case, at $1.55\,\mu\text{m}$. That is, the difference with respect to the cladding ($n_{\text{clad}} \approx 1.457$) is $\Delta n \approx 0.54$ for Si_3N_4 and $\Delta n \approx 2.02$ for Si.

Comparing the results of learning outcome 2 and Part 1, we have observed how, in order to obtain single-mode waveguides, it is necessary to work with widths smaller than $1.3\,\mu\text{m}$ (approximately) in the case of Si_3N_4, while it is necessary to go down to widths below $0.4\,\mu\text{m}$ in the case of Si. Within the single-mode range, we can check how the effective index of the guided modes (TE_0 and TM_0) is in the range of 1.48–1.61 in the case of Si_3N_4 and in the range of 1.61–2.24 in the case of Si.

Another detail worth commenting on is the sensitivity of the effective index with respect to the width. For example, for TE_0 in Si we quickly go from 1.84 to 2.64 between $0.3\,\mu\text{m}$ and $0.7\,\mu\text{m}$ respectively, which implies a rate of change of about $2.0/\mu\text{m}$. Meanwhile, in Si_3N_4, from $0.6\,\mu\text{m}$ to $1.0\,\mu\text{m}$, the effective index goes from 1.51 to only 1.58, which implies a rate of change of about $0.18/\mu\text{m}$, significantly lower than that of Si. Therefore, Si is significantly more sensitive to variations in width, a property to take into consideration.

Finally, we can compare, in the case of the deep waveguide, for TE_0, the sensitivity of the effective index with respect to the wavelength. For TE_0 in Si we quickly go from 2.14 to 1.99 between $1.5\,\mu\text{m}$ and $1.6\,\mu\text{m}$ respectively, which implies a rate of change of about $-1.5/\mu\text{m}$. Meanwhile, in Si_3N_4, over the same range, the effective index goes from 1.62 to only 1.59,

which implies a rate of change of about $-0.22/\mu m$, significantly lower in magnitude than that of Si. Therefore, Si is also significantly more sensitive to variations in wavelength. These two results begin to point in the direction that Si_3N_4 is a material that provides properties more robust against small variations, so it seems that it will be easier to work with. However, as already noted previously, Si allows us to design much more compact waveguides, less than 0.4 µm versus less than 1.3 µm wide in single-mode operation, so it is necessary to take into account all the properties and requirements to correctly assess the trade-off.

■